

```
//HUMAN INFLUENCE INDEX
```

```
var pontos = ee.FeatureCollection('projects/subtle-seat-379912/assets/Lis/tentativa2');
```

```
function bufferPoints(radius, bounds) {  
  return function(pt) {  
    pt = ee.Feature(pt);  
    return bounds ? pt.buffer(radius).bounds() : pt.buffer(radius);  
  };  
}
```

```
var pointBuffers = pontos.map(bufferPoints(10000, true));
```

```
var hiiCollection = ee.ImageCollection('projects/HII/v1/hii')  
  .filterDate('2018-01-01', '2019-01-01') //CHANGE YEAR TO COLLECT YEAR BY YEAR  
  .filterBounds(pontos);
```

```
var hiiMedian = hiiCollection.median();  
var hiiScaled = hiiMedian.divide(118);
```

```
var hiiPorPonto = hiiScaled.reduceRegions({  
  collection: pointBuffers,  
  reducer: ee.Reducer.median().combine(ee.Reducer.minMax(), null, true),  
  scale: 1000  
});
```

```
print('HII Principal Escalado (primeiros 5):', hiiPorPonto.limit(5));
```

```
print('Total de pontos:', pointBuffers.size());
```

```
var drivers = ['infrastructure', 'land_use', 'population_density', 'power', 'railways',  
'roads', 'water'];
```

```
function extractDriverStats(driverName) {  
  var driverCollection = ee.ImageCollection('projects/HII/v1/driver/' + driverName)  
    .filterDate('2001-01-01', '2019-12-31')  
    .filterBounds(pontos);  
  
  var driverMedian = driverCollection.median();  
  var driverScaled = driverMedian.divide(118);  
  
  return driverScaled.reduceRegions({  
    collection: pointBuffers,  
    reducer: ee.Reducer.median().combine(ee.Reducer.minMax(), null, true),  
    scale: 1000  
  });  
}
```

```
Export.table.toDrive({  
  collection: hiiPorPonto,  
  description: 'HII_principal_escalado_0_50_2001_2019',  
  folder: 'EarthEngine',  
  fileNamePrefix: 'HII_principal_escalado_0_50_2001_2019',  
  fileFormat: 'CSV'  
});
```

```
drivers.forEach(function(driver) {  
  var driverData = extractDriverStats(driver);  
  print('Driver ' + driver + ' escalado (primeiros 2):', driverData.limit(2));
```

```
  Export.table.toDrive({  
    collection: driverData,  
    description: 'HII_' + driver + '_escalado_0_50_2001_2019',  
    folder: 'EarthEngine',  
    fileNamePrefix: 'HII_' + driver + '_escalado_0_50_2001_2019',  
    fileFormat: 'CSV'  
  });  
});
```

```
//NDVI
```

```
var pontos = ee.FeatureCollection('projects/subtle-seat-  
379912/assets/Lis/tentativa2');
```

```
var landsat7 = ee.ImageCollection('LANDSAT/LE07/C02/T1_TOA')  
  .filterDate('2023-01-01', '2024-01-01') //CHANGE YEAR TO COLLECT YEAR BY  
YEAR  
  .filterBounds(pontos)  
  .filter(ee.Filter.lt('CLOUD_COVER', 20));
```

```
var addNDVI = function(image) {  
  var ndvi = image.normalizedDifference(['B4', 'B3']).rename('NDVI');  
  return image.addBands(ndvi);  
};
```

```
var landsatwithndvi = landsat7.map(addNDVI);  
  
var ndvimediam = landsatwithndvi.select('NDVI').median();  
  
function bufferPoints(radius, bounds) {  
  return function(pt) {  
    pt = ee.Feature(pt);  
    return bounds ? pt.buffer(radius).bounds() : pt.buffer(radius);  
  };  
}
```

```
var pointBuffers = pontos.map(bufferPoints(10000, true));
```

```
var ndviPorPonto = ndvimediam.reduceRegions({  
  collection: pointBuffers,  
  reducer: ee.Reducer.median(),  
  scale: 30,  
  tileScale: 4  
});
```

```
Export.table.toDrive({  
  collection: ndviPorPonto,  
  description: 'NDVI_por_buffer_quadrado_10km_Landsat7_2001_2024',  
  folder: 'EarthEngine',  
  fileNamePrefix: 'NDVI_por_buffer_quadrado_10km',  
  fileFormat: 'CSV'  
});
```

```
//CLIMATIC VARIABLES
```

```
var pontos = ee.FeatureCollection('projects/subtle-seat-379912/assets/tentativa2');
```

```
function bufferPoints(radius, bounds) {  
  return function(pt) {  
    pt = ee.Feature(pt);  
    return bounds ? pt.buffer(radius).bounds() : pt.buffer(radius);  
  };  
}
```

```
var pointBuffers = pontos.map(bufferPoints(10000, true));
```

```
var bio = ee.Image("WORLDCLIM/V1/BIO");
```

```
var bandasTemperatura = [  
  'bio01', 'bio02', 'bio03', 'bio04', 'bio05', 'bio06',  
  'bio07', 'bio08', 'bio09', 'bio10', 'bio11'  
];
```

```
var bandasNaoTemperatura = [  
  'bio12', 'bio13', 'bio14', 'bio15', 'bio16', 'bio17', 'bio18', 'bio19'  
];
```

```
var temperaturasCelsius = bio.select(bandasTemperatura).divide(10);
```

```
var outrasVariaveis = bio.select(bandasNaoTemperatura);
```

```
var bioConvertido = temperaturasCelsius.addBands(outrasVariaveis);
```

```
var bioPorPonto = bioConvertido.reduceRegions({  
  collection: pointBuffers,  
  reducer: ee.Reducer.median(),  
  scale: 1000  
});
```

```
Export.table.toDrive({  
  collection: bioPorPonto,  
  description: 'WorldClim_BIO_Public_Buffer10km_Celsius',  
  folder: 'EarthEngine',  
  fileNamePrefix: 'WorldClim_BIO_Public_Buffer10km_Celsius',  
  fileFormat: 'CSV'  
});
```